MAT 3007 – Optimization

Optimality Condition for Constrained Problems

Lecture 17

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Multivariable Calculus and Linear Algebra

Gradient and Hessian

Assume $f(\boldsymbol{x}) = f(x_1, x_2, \ldots, x_n)$ is continuously differentiable. Then the **gradient** of f is denoted by (an $n \times 1$ column vector):

$$\nabla f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

If f is twice continuously differentiable, then the **Hessian** of f (an $n \times n$ matrix) is given by:

$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

Example

Suppose:

$$f(x_1, x_2, x_3) := x_1^2 + x_1 x_2 + x_1 e^{x_3} + x_2 \log x_3$$

Then:

$$\nabla f(\boldsymbol{x}) = \left(2x_1 + x_2 + e^{x_3}, x_1 + \log x_3, x_1 e^{x_3} + \frac{x_2}{x_3}\right)^\top$$

and

$$\nabla^2 f(\boldsymbol{x}) = \begin{pmatrix} 2 & 1 & e^{x_3} \\ 1 & 0 & \frac{1}{x_3} \\ e^{x_3} & \frac{1}{x_3} & x_1 e^{x_3} - \frac{x_2}{x_3^2} \end{pmatrix}$$

Linear Function

Assume $f(\boldsymbol{x}) = \boldsymbol{c}^\top \boldsymbol{x}$, where $\boldsymbol{c} \in \mathbb{R}^n$ and $\boldsymbol{x} \in \mathbb{R}^n$.

Gradient:

$$\nabla f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \boldsymbol{c}$$

$$\frac{\partial f}{\partial x_i} = c_i$$

$$\frac{\partial^2 f}{\partial x_i x_j} = 0$$

Hessian:

$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} = \mathbf{0}_{n \times n}$$

Gradient of a Quadratic Function

Consider a quadratic function:

$$f(\boldsymbol{x}) = \boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x},$$



where $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{M}$ is a symmetric matrix ($\boldsymbol{M} = \boldsymbol{M}^\top$). To compute the gradient, we expand $f(\boldsymbol{x})$ as:

$$f(\boldsymbol{x}) = \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}x_i x_j.$$

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$$f(\boldsymbol{x}) = \sum_{i=1}^{n}\sum_{j=1}^{n} M_{ij}x_i x_j.$$

Using the symmetry of $\boldsymbol{M}$, we can differentiate $f(\boldsymbol{x})$ with respect to x_i and simplify:

$$\frac{\partial f}{\partial x_i} = 2\sum_{j=1}^{n} M_{ij}x_j.$$

In matrix form, this becomes:

$$\nabla f(\boldsymbol{x}) = 2\boldsymbol{M}\boldsymbol{x}.$$

Hessian of a Quadratic Function

To compute the Hessian, we start with the gradient:

$$\nabla f(\boldsymbol{x}) = 2\boldsymbol{M}\boldsymbol{x}.$$

The Hessian is defined as the matrix of second partial derivatives:

$$\nabla^2 f(\boldsymbol{x}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix} = 2\boldsymbol{M}.$$

Summary: M is symmetric

$\frac{\partial^2 f}{\partial x_i x_j} = 2M_{ij}$

- Gradient: $\nabla f(\boldsymbol{x}) = 2\boldsymbol{M}\boldsymbol{x}$.
- Hessian: $\nabla^2 f(\boldsymbol{x}) = 2\boldsymbol{M}$.

Symmetrization of Quadratic Forms

What if $\boldsymbol{M}$ is square but not symmetric? We can rewrite

$$\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x} = \frac{1}{2}\boldsymbol{x}^\top(\boldsymbol{M}+\boldsymbol{M}^\top)\boldsymbol{x}.$$

$A = \frac{1}{2}(M + M^T)$

Note: A is symmetric

$= x^T A x$

Symmetrization of Quadratic Forms

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$$\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x} = \frac{1}{2}\boldsymbol{x}^\top(\boldsymbol{M} + \boldsymbol{M}^\top)\boldsymbol{x}.$$

Explanation:

- The quadratic form $\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x}$ is a scalar (a single number).
- For any scalar, $\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x} = (\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x})^\top = \boldsymbol{x}^\top \boldsymbol{M}^\top \boldsymbol{x}$, because transposing a scalar does not change its value.
- Therefore, we can write:

$$\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x} = \frac{1}{2}\left(\boldsymbol{x}^\top \boldsymbol{M}\boldsymbol{x} + \boldsymbol{x}^\top \boldsymbol{M}^\top \boldsymbol{x}\right) = \frac{1}{2}\boldsymbol{x}^\top(\boldsymbol{M} + \boldsymbol{M}^\top)\boldsymbol{x}.$$

Definite Matrices

Definiteness: Let $\boldsymbol{A}$ be a real $n \times n$ matrix.

- $\boldsymbol{A}$ is called positive (negative) semidefinite if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} \geq 0$ (≤ 0) for all $\boldsymbol{x} \in \mathbb{R}^n$.
- The matrix $\boldsymbol{A}$ is positive (negative) definite if $\boldsymbol{x}^\top \boldsymbol{A}\boldsymbol{x} > 0$ (< 0) for all $\boldsymbol{x} \in \mathbb{R}^n \backslash \{\boldsymbol{0}\}$.
- $\boldsymbol{A}$ is said to be indefinite if $\boldsymbol{A}$ is neither positive semidefinite nor negative semidefinite.

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Theorem: Eigenvalues and Definiteness

Let $\boldsymbol{A}$ be a real symmetric $n \times n$ matrix with eigenvalues $\lambda_i \in \mathbb{R}$, $i = 1, \ldots, n$. It follows:

- $\boldsymbol{A}$ is positive (negative) semidefinite iff $\lambda_i \geq 0$ (≤ 0) for all i.
- $\boldsymbol{A}$ is positive (negative) definite iff $\lambda_i > 0$ (< 0) for all i.
- $\boldsymbol{A}$ is indefinite iff there are $i, j \in \{1, \ldots, n\}$ with $\lambda_i > 0$ and $\lambda_j < 0$.

Recap: Checking Definiteness

How can we easily check definiteness of a 2×2 matrix $\boldsymbol{A}$?

- It holds that $\mathrm{tr}(\boldsymbol{A}) = \lambda_1 + \lambda_2$ and $\det(\boldsymbol{A}) = \lambda_1 \lambda_2$.

Recap: Checking Definiteness

How can we easily check definiteness of a 2×2 matrix $\boldsymbol{A}$?

- It holds that $\mathrm{tr}(\boldsymbol{A}) = \lambda_1 + \lambda_2$ and $\det(\boldsymbol{A}) = \lambda_1 \lambda_2$.
- If $\det(\boldsymbol{A}) < 0$: The eigenvalues have different signs. $\boldsymbol{A}$ must be indefinite.
- If $\det(\boldsymbol{A}) = 0$: One of the eigenvalues is zero. If $\mathrm{tr}(\boldsymbol{A}) \geq 0$, then $\boldsymbol{A}$ is positive semidefinite. If $\mathrm{tr}(\boldsymbol{A}) \leq 0$, then $\boldsymbol{A}$ is negative semidefinite.
- If $\det(\boldsymbol{A}) > 0$: Both eigenvalues have the same sign. If $\mathrm{tr}(\boldsymbol{A}) > 0$, then $\boldsymbol{A}$ is positive definite. If $\mathrm{tr}(\boldsymbol{A}) < 0$, then $\boldsymbol{A}$ is negative definite.

Recap

First-Order Necessary Conditions (FONC)

First-Order Necessary Conditions

If $\boldsymbol{x}^*$ is a local minimizer of the unconstrained problem $\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$, then we must have $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$.

First-Order Necessary Conditions

If $\boldsymbol{x}^*$ is a local minimizer of the unconstrained problem $\min_{\boldsymbol{x}\in\mathbb{R}^n} f(\boldsymbol{x})$, then we must have $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$.

Why? If $\nabla f(\boldsymbol{x}) \neq \mathbf{0}$, then we can find a vector $\boldsymbol{d}$ with $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$. Therefore, by Taylor expansion:

$$f(\boldsymbol{x} + t\boldsymbol{d}) = f(\boldsymbol{x}) + t\nabla f(\boldsymbol{x})^\top \boldsymbol{d} + o(t), \quad t \to 0.$$

By choosing t small enough, we can find a point $\bar{\boldsymbol{x}} = \boldsymbol{x} + t\boldsymbol{d}$ in the neighborhood of $\boldsymbol{x}$ such that $f(\bar{\boldsymbol{x}}) < f(\boldsymbol{x})$.

Second-Order Necessary Condition (SONC)

Theorem: Second-Order Necessary Conditions

If f is twice continuously differentiable and $\boldsymbol{x}^*$ is a local minimizer of f, then:

1. $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$;
2. $\nabla^2 f(\boldsymbol{x}^*)$ is positive semidefinite.

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Why? Consider the 2nd-order Taylor expansion

$$f(\boldsymbol{x} + t\boldsymbol{d}) = f(\boldsymbol{x}) + t\nabla f(\boldsymbol{x})^\top \boldsymbol{d} + \frac{1}{2}t^2 \boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x})\boldsymbol{d} + o(t^2).$$

When the first-order necessary condition holds, we have:

$$f(\boldsymbol{x} + t\boldsymbol{d}) = f(\boldsymbol{x}) + \frac{1}{2}t^2 \boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x})\boldsymbol{d} + o(t^2).$$

If $\boldsymbol{x}$ is a local minimizer, we also need $\boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x})\boldsymbol{d}$ to be nonnegative for every $\boldsymbol{d} \in \mathbb{R}^n$.

Second-Order Sufficient Condition (SOSC)

Theorem: Second-Order Sufficient Conditions

Let f be twice continuously differentiable. If $\boldsymbol{x}^*$ satisfies:

1. $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$;
2. $\nabla^2 f(\boldsymbol{x}^*)$ is positive definite?

then $\boldsymbol{x}^*$ is a strict local minimum of f.

Second-Order Sufficient Condition (SOSC)

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Why? Consider the 2nd-order Taylor expansion

$$f(\boldsymbol{x}+\boldsymbol{d}) = f(\boldsymbol{x}) + \underbrace{\nabla f(\boldsymbol{x})^\top \boldsymbol{d}}_{=0} + \frac{1}{2}\boldsymbol{d}^\top \nabla^2 f(\boldsymbol{x})\boldsymbol{d} + o(\|\boldsymbol{d}\|^2).$$

When $\nabla^2 f(\boldsymbol{x}^*)$ is positive definite, we have $\boldsymbol{d}^T \nabla^2 f(\boldsymbol{x}^*)\boldsymbol{d} \geq \mu \|\boldsymbol{d}\|^2$ for all $\boldsymbol{d}$ with $\mu := \lambda_{\min}(\nabla^2 f(\boldsymbol{x}^*)) > 0$.

For Maximization Problems

Theorem: FONC for Maximization

If $\boldsymbol{x}^*$ is a local (unconstrained) maximizer of f, then we must have $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$.

Theorem: SONC for Maximization

If $\boldsymbol{x}^*$ is a local maximizer of f, then we must have 1.) $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$; 2.) $\nabla^2 f(\boldsymbol{x}^*)$ is negative semidefinite.

Theorem: SOSC for Maximization

If $\boldsymbol{x}^*$ satisfies 1.) $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$; 2.) $\nabla^2 f(\boldsymbol{x}^*)$ is negative definite, then $\boldsymbol{x}^*$ is a strict local maximizer.

Saddle Points

Indefiniteness and Saddle Points

Definition: Stationary Points and Saddle Points

- A point $\boldsymbol{x}$ satisfying $\nabla f(\boldsymbol{x}) = \mathbf{0}$ is called critical point or stationary point.
- A stationary point is called saddle point if it is neither a local minimizer nor a local maximizer.

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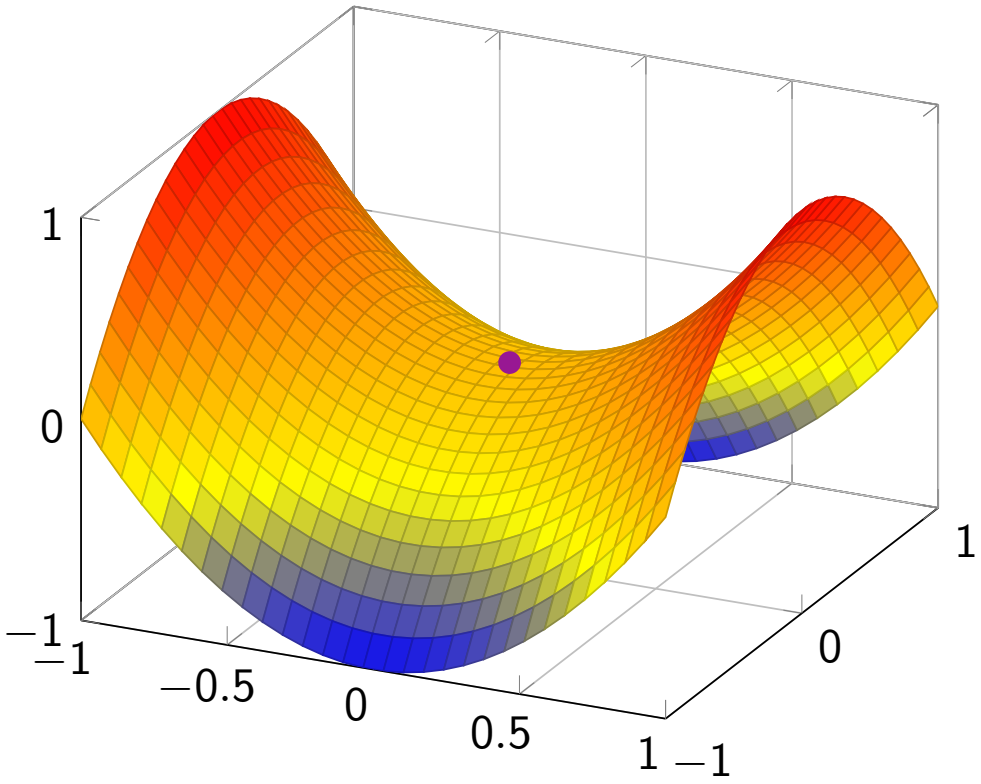
Corollary: A Sufficient Condition for Saddle Points

Suppose that $\boldsymbol{x}^*$ is a stationary point and that the Hessian $\nabla^2 f(\boldsymbol{x}^*)$ is indefinite, then $\boldsymbol{x}^*$ is a saddle point.

Not a necessary condition. e.g. $f(x) = x^3$

why? Because some eigenvalues are positive some are negative, it does not satisfy SONC for local minimizer/maxmizer
→ It has to be a saddle point.

Example: Saddle Point



Plot of the function $f(\boldsymbol{x}) = x_1^2 - x_2^2$.

- The gradient is $\nabla f(\boldsymbol{x}) = (2x_1, -2x_2)^\top$ and $\boldsymbol{x}^* = (0,0)^\top$ is the only stationary point of f. Since $\nabla^2 f(\boldsymbol{x}^*)$ is indefinite, $\boldsymbol{x}^*$ has to be a saddle point.

Examples

Examples — I

$$f(x) := x^4 - 9x^2 + 4x - 1$$

$x_1 = -1 - \sqrt{6}/2$, $x_2 = -1 + \sqrt{6}/2$ and $x_3 = 2$ satisify FONC. The second derivative is

$$f''(x) = 12x^2 - 18.$$

$f''(x_2) < 0$

$x_2 = -1 + \sqrt{6}/2$ is a strict local maximizer. $x_1 = -1 - \sqrt{6}/2$ and $x_3 = 2$ are strict local minimizer.

$f''(x_1) > 0$

$f''(x_3) > 0$

Examples — II

the least squares problem:

$$\text{minimize}_{\boldsymbol{\beta}} \quad \|\boldsymbol{X}\boldsymbol{\beta} - \boldsymbol{y}\|^2 = \boldsymbol{\beta}^\top \boldsymbol{X}^\top \boldsymbol{X}\boldsymbol{\beta} - 2\boldsymbol{\beta}^\top \boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{y}$$

Examples — II

the least squares problem: quadratic and X^TX is symmetric.

$$\text{minimize}_{\beta} \quad \|\boldsymbol{X}\beta - \boldsymbol{y}\|^2 = \beta^\top \boldsymbol{X}^\top \boldsymbol{X}\beta - 2\beta^\top \boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{y}$$

The FONC for the least squares problem is:

$$\boldsymbol{X}^\top \boldsymbol{X}\beta - \boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{0}.$$

Examples — II

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$$\text{minimize}_{\beta} \quad \|\boldsymbol{X}\beta - \boldsymbol{y}\|^2 = \beta^\top \boldsymbol{X}^\top \boldsymbol{X}\beta - 2\beta^\top \boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{y}$$

The FONC for the least squares problem is:

$$\boldsymbol{X}^\top \boldsymbol{X}\beta - \boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{0}.$$

Hessian $= 2X^TX$

Note: X^TX is always positive semi-definite.

If $\boldsymbol{X}^\top \boldsymbol{X}$ is positive definite, then the solution β of the FONC

$$\boldsymbol{X}^\top \boldsymbol{X}\beta = \boldsymbol{X}^\top \boldsymbol{y}$$

$\beta = (X^TX)^{-1}X^Ty$

is unique and it satisfies the second-order sufficient conditions.

$\rightsquigarrow$ It must be the unique global minimizer of the problem.

Example — II

$$\min_{\boldsymbol{x}\in\mathbb{R}^2} \; f(\boldsymbol{x}) = x_1^2 x_2 + x_1 x_2^3 - 5x_1 x_2$$

Find all local minimizer, local maximizer, and saddle points of f.

$$\nabla f(x) = \begin{bmatrix} 2x_1x_2 + x_2^3 - 5x_2 \\ x_1^2 + 3x_1x_2^2 - 5x_1 \end{bmatrix} = \begin{bmatrix} x_2(2x_1 + x_2^2 - 5) \\ x_1(x_1 + 3x_2^2 - 5) \end{bmatrix} = 0$$

(a) $x_1 = x_2 = 0$

(b) $x_1 = 0$ and $2x_1 + x_2^2 - 5 = 0$

$\Leftrightarrow \begin{cases} x_1 = 0 \\ x_2 = \pm\sqrt{5} \end{cases}$

(c) $x_2 = 0$ and $x_1 + 3x_2^2 - 5 = 0$

$\Leftrightarrow \begin{cases} x_2 = 0 \\ x_1 = 5 \end{cases}$

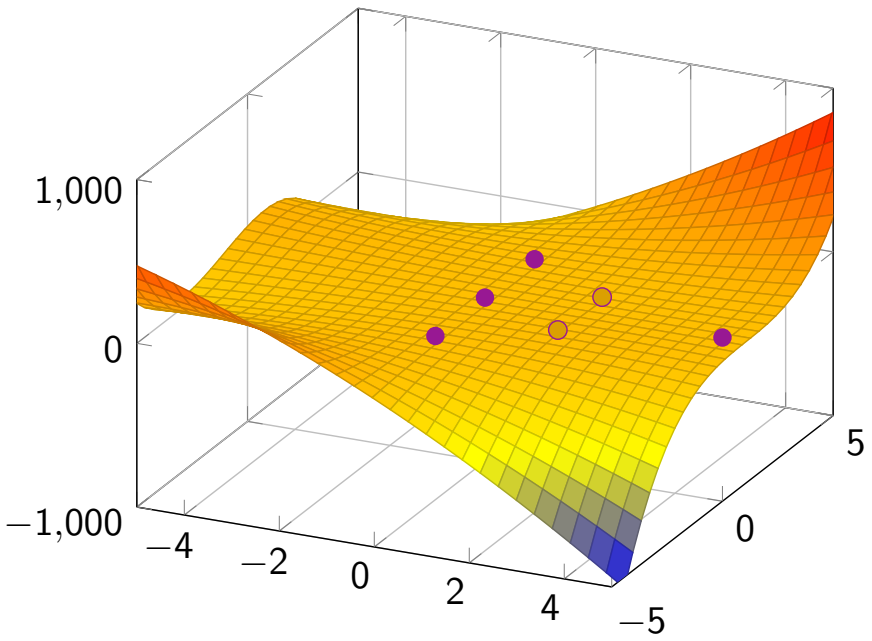
(d) $2x_1 + x_2^2 - 5 = 0$ and $x_1 + 3x_2^2 - 5 = 0$

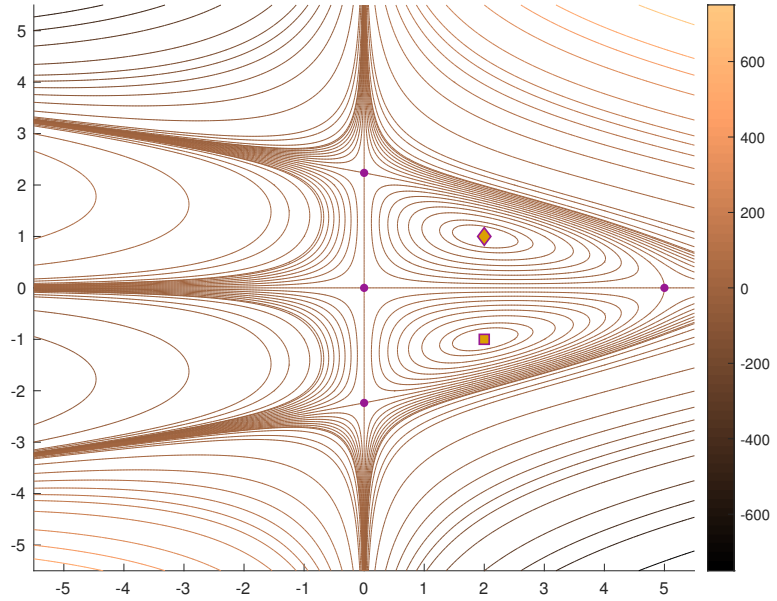
$\Leftrightarrow \begin{cases} x_1 = 2 \\ x_2 = \pm 1 \end{cases}$

Example — II: Continued

$\boldsymbol{x}^*$	$f(\boldsymbol{x}^*)$	$\nabla^2 f(\boldsymbol{x}^*)$	$\lambda_1,\ \lambda_2$	Definiteness	Conclusion
$\begin{pmatrix}0\\0\end{pmatrix}$		$\begin{pmatrix}0 & -5\\-5 & 0\end{pmatrix}$	$5,\ -5$	indef	saddle point
$\begin{pmatrix}5\\0\end{pmatrix}$		$\begin{pmatrix}0 & 5\\5 & 0\end{pmatrix}$	$5,\ -5$	indef	saddle ..
$\begin{pmatrix}0\\\sqrt{5}\end{pmatrix}$		$\begin{pmatrix}\sqrt{5} & 10\\10 & 0\end{pmatrix}$	$5\sqrt{5},\ -4\sqrt{5}$	indef	saddle ..
$\begin{pmatrix}0\\-\sqrt{5}\end{pmatrix}$		$\begin{pmatrix}-\sqrt{5} & 10\\10 & 0\end{pmatrix}$	$4\sqrt{5},\ -5\sqrt{5}$	indef	saddle ..
$\begin{pmatrix}2\\1\end{pmatrix}$		$\begin{pmatrix}1 & 2\\2 & 12\end{pmatrix}$	$12.4,\ 0.7$	positive def.	strict local min
$\begin{pmatrix}2\\-1\end{pmatrix}$		$\begin{pmatrix}-1 & 2\\2 & -12\end{pmatrix}$	$-0.7,\ -12.4$	negative def	strict local max

Example — II: Continued





Optimality Conditions for Constrained Problems

Consider optimization problems of the form:

General Nonlinear Optimization Problem:

$$\begin{aligned} \text{minimize}_{\boldsymbol{x}\in\mathbb{R}^n} \quad & f(\boldsymbol{x}) \\ \text{subject to} \quad & g_i(\boldsymbol{x}) \le 0, \quad \forall\ i = 1,\dots,m, \\ & h_j(\boldsymbol{x}) = 0, \quad \forall\ j = 1,\dots,p. \end{aligned} \tag{1}$$

- The feasible set is $\Omega = \{\boldsymbol{x} \in \mathbb{R}^n : g(\boldsymbol{x}) \le \mathbf{0},\ h(\boldsymbol{x}) = \mathbf{0}\}$.
- f, g_i, and h_j are continuously differentiable for all i and j.

Our Goal:

- Derive optimality conditions for such class of problems.

Constrained Problems

We have derived necessary and sufficient conditions for the local minimum for unconstrained problems.

- What is the difference between constrained and unconstrained problems?

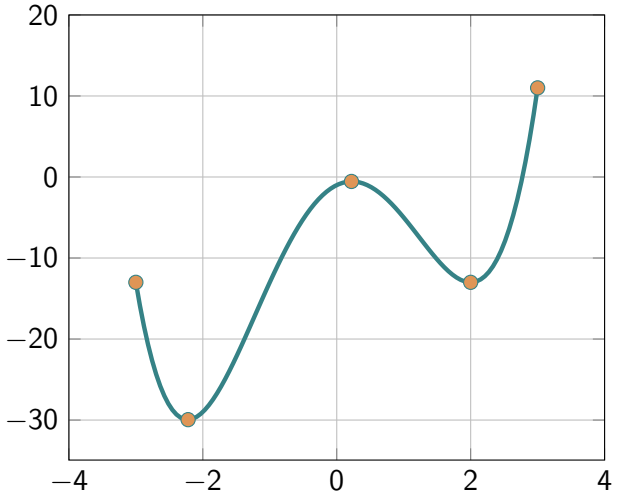
Constrained Problems

We have derived necessary and sufficient conditions for the local minimum for unconstrained problems.

- What is the difference between constrained and unconstrained problems?

Let us consider the constrained problem

$$\min_{x\in\mathbb{R}} f(x) := x^4 - 9x^2 + 4x - 1 \quad \text{subject to} \quad x \in [-3, 3].$$



In addition to the original local maximizer ($x_2 = \sqrt{6}/2 - 1$), there are two more local maximizer at the boundary ($x = \pm 3$).

At the boundary ($x = 3$), the FONC is not satisfied:

$$f'(3) = 58 > 0.$$

However, at this point, in order to stay feasible, we can only go leftward.

Thus, $f(x + d) < f(x)$ in a small neighborhood of x in the feasible region. Hence, $x = 3$ is a local maximizer.

$\rightsquigarrow$ It means that the developed FONC ($\nabla f(\boldsymbol{x}) = \mathbf{0}$) for unconstrained problems is not enough for constrained problems.

Feasible Directions

We now formalize the above arguments.

Definition: Feasible Direction

Given $\boldsymbol{x} \in \Omega$, we call $\boldsymbol{d}$ a feasible direction at $\boldsymbol{x}$ if there exists $\bar{t} > 0$ such that $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all $0 \leq t \leq \bar{t}$.

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Example:

- If $\Omega = \{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}\}$, then all feasible directions at $\boldsymbol{x}$ are given by $\{\boldsymbol{d} : \boldsymbol{A}\boldsymbol{d} = \boldsymbol{0}\}$.
- If $\Omega = \{\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}\}$, then the feasible directions at $\boldsymbol{x}$ are given by $\{\boldsymbol{d} : \boldsymbol{a}_i^\top \boldsymbol{d} \leq 0$ if $\boldsymbol{a}_i^\top \boldsymbol{x} = b_i\}$.

for all i with

<A> $\Omega = \{x : Ax = b\}$. d is a feasible direction iff

$\exists \bar{t}$ s.t. $A(x+td) = b$ for $t \in [0, \bar{t}]$.

$\Leftrightarrow$ ---- $\underline{Ax} + tAd = b$ ----

$= 0$ because $x \in \Omega$

$\Leftrightarrow$ ----- $tAd = 0$ ----

$\Leftrightarrow$ $Ad = 0$

<B> $S = \{x : Ax \le b\}$ $\quad A = \begin{bmatrix} — & a_1^T & — \\ — & a_2^T & — \\ & \vdots & \\ — & a_n^T & — \end{bmatrix}$

d is a feasible direction iff

$\exists \bar{t}$ s.t $A(x+td) \le b$ for $t \in [0, \bar{t}]$

$\Leftrightarrow$ ------ $a_i^T x + t a_i^T d \le b_i \ \forall i$ ------

① consider i with $a_i^T x < b_i$. Then

$a_i^T x + t a_i^T d \le b_i$ is satisfied as long as t is small enough

② Consider i with $a_i^T x = b_i$. Then

$a_i^T x + t a_i^T d \le b_i \Leftrightarrow t a_i^T d \le 0$

is satified when $a_i^T d \le 0$.

To conclude, the set of feasible direction is

$\{d : a_i^T d \le 0$ for all i with $a_i^T x = b_i\}$

Theorem: FONC for Constrained Problems

Let $\boldsymbol{x}^*$ be a local minimum of $\min_{\boldsymbol{x}\in\Omega} f(\boldsymbol{x})$. Then for any feasible direction $\boldsymbol{d}$ at $\boldsymbol{x}^*$, we must have $\nabla f(\boldsymbol{x}^*)^\top \boldsymbol{d} \geq 0$.

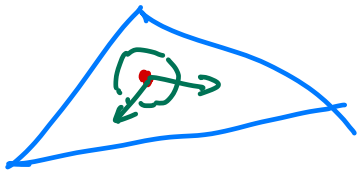
Proof:

$\rightsquigarrow$ As before by Taylor expansion and using $\boldsymbol{x} + t\boldsymbol{d} \in \Omega$ for all t sufficiently small.

Remark:

- In the unconstrained case, all directions are feasible. Thus, we must have $\nabla f(\boldsymbol{x}^*) = \boldsymbol{0}$.

$\rightsquigarrow$ This is also true, when $\boldsymbol{x}^*$ lies in the interior of Ω.



An Alternative View: Descent Directions

Definition: Descent Direction

Let f be continuously differentiable. Then $\boldsymbol{d}$ is called a descent direction at $\boldsymbol{x}$ if and only if $\nabla f(\boldsymbol{x})^\top \boldsymbol{d} < 0$.

Remark:

- If $\boldsymbol{d}$ is a descent direction at $\boldsymbol{x}$, then there exists $\bar{\alpha} > 0$ such that $f(\boldsymbol{x} + \alpha \boldsymbol{d}) < f(\boldsymbol{x})$ for all $0 < \alpha \leq \bar{\alpha}$.

If we denote the set of feasible directions at $\boldsymbol{x}$ by $S_\Omega(\boldsymbol{x})$ and the set of descent directions at $\boldsymbol{x}$ by $S_D(\boldsymbol{x})$, then the first order necessary condition can be written as:

$$S_\Omega(\boldsymbol{x}^*) \cap S_D(\boldsymbol{x}^*) = \emptyset$$

$\rightsquigarrow$ There are no feasible descent directions.

Today's lecture

- FONC for Constrained Problems

Next lecture

- KKT Condition

Questions?